

MIAE 221: Materials Science for Engineers

Lecture 7: Defects — Imperfections in Solids & Dislocations (Explained)

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Teacher Material: Lecture 7 (Prof. Mamoun Medraj) · Correlated with Callister Chapter 4

Executive Overview & Core Concepts

No engineering material is crystalline perfection; real materials contain structural imperfections that govern mechanical strength, electrical resistivity, semiconductor doping, and plastic deformation. This explained lecture guide covers:

1. **0D Point Defects:** Vacancies, self-interstitials, and the thermodynamic necessity of equilibrium vacancy concentrations.
 2. **Arrhenius Thermodynamics & Activation Energy (Q_v):** Mathematical formulation and Arrhenius slope analysis for vacancy quantification.
 3. **Solid Solutions:** Substitutional and interstitial solid solution mechanisms.
 4. **Hume-Rothery Rules:** The four empirical criteria governing solid solubility limits.
 5. **1D Linear Defects (Dislocations):** Edge, screw, and mixed dislocations, the Burgers vector (**b**), and dislocation glide physics.
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1. Zero-Dimensional Point Defects

Point defects are localized lattice disruptions involving one or two atomic diameters:

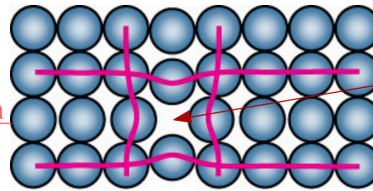


Point Defects

vacancy → a lattice site that is missing an atom:

All crystals contain some vacancies.

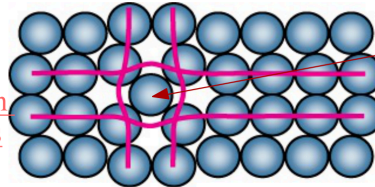
distortion of planes



Vacancy

self-interstitial → an atom from the crystal that crowds its way into an otherwise empty void between atoms

distortion of planes



self-interstitial

*Self-interstitials are far less common than vacancies because of the relatively **energy** required to squeeze an atom into the small voids between existing sites.*

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Figure 7.1: Zero-dimensional point defects in crystalline solids (Dr. Medraj MIAE 221 Lecture 7, Slide 4 & Callister Fig. 4.1). (a) Vacancy: an unoccupied atomic site causing inward compressive lattice distortion. (b) Self-interstitial: an extra atom squeezed into an interstitial site, causing severe outward tensile lattice strain.

1. Vacancy

- An atomic site that is normally occupied in the perfect crystal lattice, but is missing an atom.
- Produced during solid-state crystallization, rapid quenching, or high-energy radiation damage.
- Causes adjacent surrounding atoms to relax inward, creating localized compressive strain fields.

2. Self-Interstitial

- A host atom squeezed into an interstitial void (a small empty space between lattice atoms) that is not normally occupied.
- Since the diameter of the host atom is substantially larger than the interstitial void space, introducing a self-interstitial causes **severe localized lattice strain**.
- **Thermodynamic Consequence:** The formation energy of a self-interstitial ($Q_i \approx 3 - 5$ eV) is far higher than that of a vacancy ($Q_v \approx 0.8 - 1.2$ eV). Consequently, the equilibrium concentration of self-interstitials is negligible under ordinary thermal conditions ($N_i/N \approx 10^{-30}$ at room temperature).

2. Equilibrium Vacancy Concentration & Arrhenius Thermodynamics

Why Vacancies Exist in Thermodynamic Equilibrium

Unlike dislocations and grain boundaries (which are non-equilibrium defects), vacancies are **thermodynamically stable equilibrium defects**:

- Creating a vacancy requires an enthalpy expenditure ($\Delta H_v > 0$) to break interatomic bonds.
- However, distributing vacancies randomly throughout the lattice introduces spatial disorder, dramatically increasing the **configurational entropy** ($\Delta S_v > 0$).
- At any temperature $T > 0$ K, the Gibbs free energy of the crystal ($G = H - TS$) is minimized at a specific, non-zero concentration of vacancies:

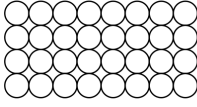


Equilibrium Concentration: Point Defects

- Equilibrium concentration varies with temperature!

No. of defects

No. of potential defect sites.



Each lattice site is a potential vacancy site

$$\frac{N_D}{N} = \exp \left[\frac{-Q_D}{kT} \right]$$

Activation energy

Temperature

Boltzmann's constant

(1.38×10^{-23} J/atom K)

(8.62×10^{-5} eV/atom K)

This is called an *Arrhenius* equation, after Svante August Arrhenius, a Swedish chemist who won the *Nobel* prize in chemistry in 1903.

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Figure 7.2: Thermodynamic equilibrium vacancy concentration curve (Dr. Medraj Lecture 7, Slide 5 & Callister Fig. 4.2). The vacancy fraction increases exponentially with absolute temperature according to the Boltzmann distribution.

The Governing Arrhenius Equation

$$\frac{N_v}{N} = \exp \left(-\frac{Q_v}{k_B T} \right)$$

Where:

- N_v = Number of equilibrium vacancies per unit volume (m^{-3} or cm^{-3}).
- N = Total number of atomic lattice sites per unit volume (m^{-3} or cm^{-3}).
- Q_v = Activation energy required to form one vacancy (J/atom or eV/atom).
- k_B = Boltzmann constant:
 - $k_B = 1.38 \times 10^{-23}$ J/(atom · K) (use when Q_v is given in Joules).
 - $k_B = 8.62 \times 10^{-5}$ eV/(atom · K) (use when Q_v is given in electron-volts).
- T = Absolute temperature in **Kelvin** ($T(\text{K}) = T(^{\circ}\text{C}) + 273.15$).

Step-by-Step Calculation: Total Atomic Sites (N)

To determine the absolute number of vacancies per cubic meter (N_v), one must first compute the total number of atomic sites N per cubic meter from the material's bulk density (ρ), molar mass (A), and Avogadro's number ($N_A = 6.022 \times 10^{23}$ atoms/mol):

$$N = \frac{\rho \times N_A}{A}$$

Dimensional Verification

:

$$N = \frac{[\text{g/m}^3] \times [\text{atoms/mol}]}{[\text{g/mol}]} = \text{atoms/m}^3$$

Comprehensive Solved Problem: Vacancy Concentration in Copper (Dr. Medraj Lecture 7 Example)

Problem Statement: Calculate the equilibrium number of vacancies per cubic meter (N_v) in pure copper at 1000°C. Given data:

- Activation energy for vacancy formation: $Q_v = 0.90$ eV/atom
- Atomic weight of copper: $A_{Cu} = 63.55$ g/mol
- Density of copper at 1000°C: $\rho = 8.40$ g/cm³

Step-by-Step Solution:

1. Step 1: Convert Temperature to Absolute Kelvin:

$$T = 1000^\circ\text{C} + 273.15 = \mathbf{1273.15\text{ K}}$$

2. Step 2: Convert Density to SI Units (g/m³):

$$\rho = 8.40 \text{ g/cm}^3 \times \left(\frac{100 \text{ cm}}{1 \text{ m}} \right)^3 = 8.40 \times 10^6 \text{ g/m}^3$$

3. Step 3: Compute Total Atomic Lattice Sites (N):

$$N = \frac{\rho N_A}{A_{Cu}} = \frac{(8.40 \times 10^6 \text{ g/m}^3)(6.022 \times 10^{23} \text{ atoms/mol})}{63.55 \text{ g/mol}}$$
$$N = \frac{5.0585 \times 10^{30}}{63.55} = \mathbf{7.96 \times 10^{28} \text{ atoms/m}^3}$$

4. Step 4: Compute the Exponential Arrhenius Argument: Using $k_B = 8.62 \times 10^{-5}$ eV/K:

$$\frac{Q_v}{k_B T} = \frac{0.90 \text{ eV}}{(8.62 \times 10^{-5} \text{ eV/K})(1273.15 \text{ K})} = \frac{0.90}{0.109745} = 8.2008$$

5. Step 5: Compute the Equilibrium Vacancy Fraction (N_v/N):

$$\frac{N_v}{N} = \exp(-8.2008) = e^{-8.2008} = \mathbf{2.744 \times 10^{-4}}$$

(Physical interpretation: at 1000°C, approximately 1 out of every 3600 lattice sites in copper is empty).

6. Step 6: Compute Absolute Number of Vacancies (N_v):

$$N_v = N \times \left(\frac{N_v}{N} \right) = (7.96 \times 10^{28} \text{ m}^{-3}) \times (2.744 \times 10^{-4})$$

$$\mathbf{N_v = 2.18 \times 10^{25} \text{ vacancies/m}^3}$$

Measuring Activation Energy via Arrhenius Plots

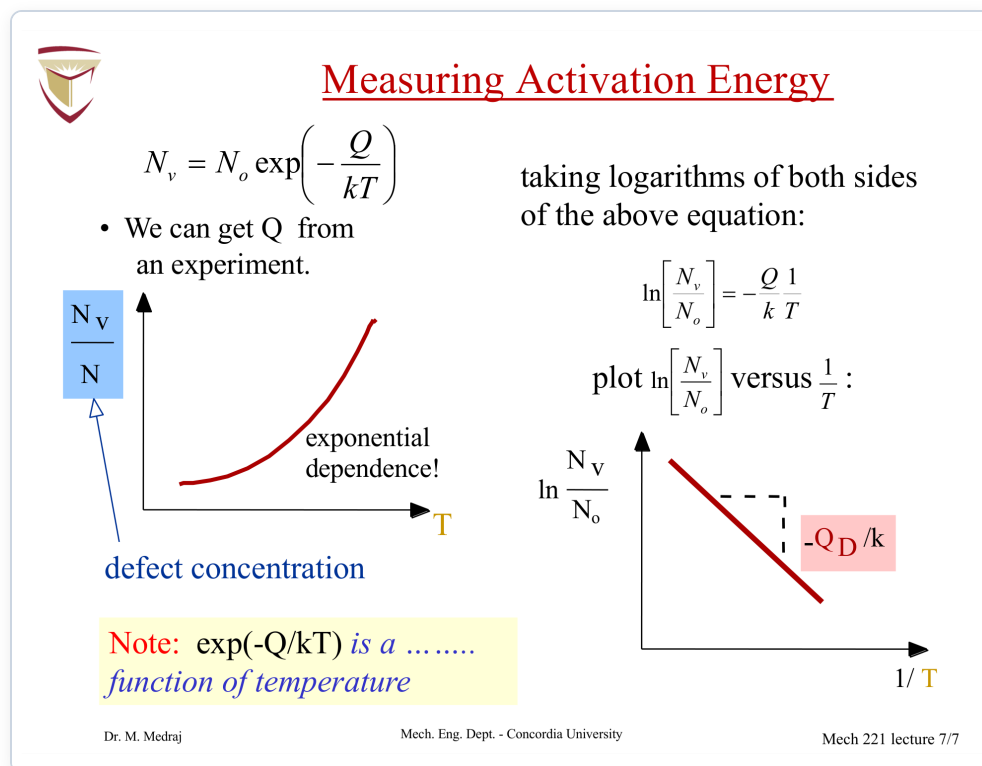


Figure 7.3: Arrhenius plot of $\ln(N_v/N)$ versus inverse absolute temperature $1/T$ (Dr. Medraj Lecture 7, Slide 8 & Callister Fig. 4.3).
The slope of the resulting linear curve yields the activation energy $-Q_v/k_B$.

Taking the natural logarithm of both sides of the Arrhenius equation:

$$\ln\left(\frac{N_v}{N}\right) = \ln\left[\exp\left(-\frac{Q_v}{k_B T}\right)\right] = -\frac{Q_v}{k_B} \left(\frac{1}{T}\right)$$

This has the linear form $y = mx + b$, where:

- $y = \ln(N_v/N)$
- $x = \frac{1}{T}$
- Slope $m = -\frac{Q_v}{k_B}$**

By measuring the vacancy concentration at two temperatures (T_1 and T_2):

$$\ln \left(\frac{N_{v1}}{N} \right) - \ln \left(\frac{N_{v2}}{N} \right) = -\frac{Q_v}{k_B} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$Q_v = -k_B \frac{\ln(N_{v1}/N_{v2})}{\frac{1}{T_1} - \frac{1}{T_2}}$$

3. Impurities in Solids & Solid Solutions

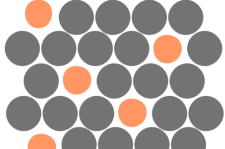
A pure elemental metal does not exist commercially; all engineering alloys consist of a **solvent** (host matrix) containing **solute** (minority guest atoms).



Point Defects in Alloys

Two outcomes if impurity (B) added to host (A):

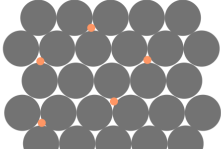
- **Solid solution** of B in A (i.e., random dist. of point defects)



Substitutional solid soln.

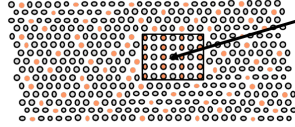
(e.g., **Cu** in Ni)

OR



Interstitial solid soln.

(e.g., **C** in Fe)
- Solid solution of B in A plus particles of a new phase (usually for a larger amount of B)



Second phase particle

 - *different*
 - *often different*

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Figure 7.4: Atomic mechanisms of solid solution formation (Dr. Medraj Lecture 7, Slide 9 & Callister Fig. 4.4). (a) Substitutional solid solution: solute atoms replace host solvent atoms in lattice positions. (b) Interstitial solid solution: small solute atoms occupy interstitial voids between host atoms.

1. Substitutional Solid Solutions

Solute atoms directly replace host atoms at regular lattice positions:

- Example
: Copper-Nickel (Cu-Ni) brass/monel alloys. Nickel atoms occupy regular FCC copper lattice sites.

2. Interstitial Solid Solutions

Solute atoms occupy the tiny empty spaces (interstitial sites) between the host atoms:

- Solute atoms must be substantially smaller than host atoms.
- The atomic radius ratio must satisfy:

$$\frac{r_{\text{solute}}}{r_{\text{solvent}}} < 0.59$$

-

Engineering Example

: Carbon in Iron (C-Fe steel). Carbon ($r = 0.071 \text{ nm}$) fits interstitially into Iron ($r = 0.124 \text{ nm}$).

- **Solubility Limits:** In BCC α -ferrite, interstitial sites are small and distorted; maximum carbon solubility is only 0.022 wt at 727°C. In FCC γ -austenite, the octahedral interstitial void is larger; carbon solubility reaches 2.14 wt.

4. The Hume-Rothery Rules for Substitutional Solid Solubility

For two metals to exhibit high or complete solid solubility in each other (e.g., forming an isomorphous phase diagram like Cu-Ni), they must satisfy the **four Hume-Rothery Rules**:



Solid Solution

Not all elements form solid solutions. There are very specific rules that govern the extent to which solid solutions can form.

For appreciable substitutional solubility to occur the following factors must hold:

- **Atomic size factor**

the difference in atomic radii between the two atoms must be $\leq \dots\dots\dots$

- **Electronegativity**

the difference in electronegativity between the two atoms must be $\leq \pm 0.4 \text{ eV}$ (i.e. large differences $\rightarrow$ compound formation (intermetallics))

- **Crystal structure**

The crystal structure of each element must be the $\dots\dots\dots$

- **Valence**

For a given solvent, a solute with a higher valency is more likely to be soluble than one of lower valency, all other factors being equal.

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Figure 7.5: Summary of the 4 Hume-Rothery empirical criteria for substitutional solid solubility (Dr. Medraj Lecture 7, Slide 11 & Callister Table 4.1).

Rule Number	Hume-Rothery Criterion	Physical Justification & Numerical Limit
Rule 1: Atomic Size Factor	The difference in atomic radii (Δr) must be less than 15	$\Delta r = \frac{ r_{\text{solute}} - r_{\text{solvent}} }{r_{\text{solvent}}} \times 100$. If $\Delta r > 15$, lattice strain is too severe, forcing phase separation.
Rule 2: Crystal Structure Match	Both elements must possess the same crystal structure	E.g., both FCC, both BCC, or both HCP. Mismatched lattices cannot form an unbroken series of solid solutions across all compositions.
Rule 3: Electronegativity Proximity	Electronegativities must be comparable	If the electronegativity difference ΔX is large, the elements form a brittle intermetallic compound rather than a solid solution.
Rule 4: Valency Rule	A metal dissolves another metal of higher valence more readily than one of lower valence	Maximum solubility occurs when solute valency $\geq$ solvent valency.

Case Study: Cu-Ni Complete Solid Solubility

- $r_{Cu} = 0.128 \text{ nm}$, $r_{Ni} = 0.125 \text{ nm} \implies \Delta r = \frac{0.128 - 0.125}{0.128} = 2.3$ (Passes Rule 1).
- Crystal structure: Both are FCC (Passes Rule 2).
- Electronegativities: $X_{Cu} = 1.9$, $X_{Ni} = 1.8 \implies \Delta X = 0.1$ (Passes Rule 3).
- Valences: Cu^{+1}/Cu^{+2} , Ni^{+2} (Passes Rule 4).
- **Result:** Copper and Nickel form complete, continuous solid solutions across all proportions from 0 to 100 Ni.

5. One-Dimensional Linear Defects: Dislocations

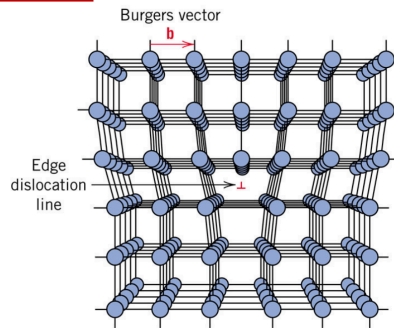
A **dislocation** is a one-dimensional (linear) crystalline defect around which atoms are misaligned. Dislocations are the fundamental carriers of **plastic deformation (ductility)** in crystalline metals.

1. Edge Dislocation



Edge Dislocation

- Think of edge dislocation as an extra half-plane of atoms inserted in a crystal.
- Misalignment of atomic planes due to the extra half plane.



Burger's vector (b**)** = magnitude + direction of lattice distortion.

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Figure 7.6: Atomic configuration of an edge dislocation (Dr. Medraj Lecture 7, Slide 16 & Callister Fig. 4.7). An extra half-plane of atoms terminates at line **t**. Above the line, atoms are in compression; below, they are in tension. The Burgers vector **b** is strictly perpendicular to the dislocation line: $\mathbf{b} \perp \mathbf{t}$.

- **Structure:** Formed by introducing an **extra half-plane of atoms** that terminates within the crystal.
- **Dislocation Line (**t**):** The line running along the bottom edge of the extra half-plane.
- **Strain Fields:**
 - Region **above** the slip plane: Atoms are squeezed together $\implies$ localized **compressive strain**.
 - Region **below** the slip plane: Atoms are pulled apart $\implies$ localized **tensile strain**.
- **Burgers Vector Relationship:**

$$\mathbf{b} \perp \mathbf{t} \quad (\text{Burgers vector is PERPENDICULAR to dislocation line})$$

2. Screw Dislocation

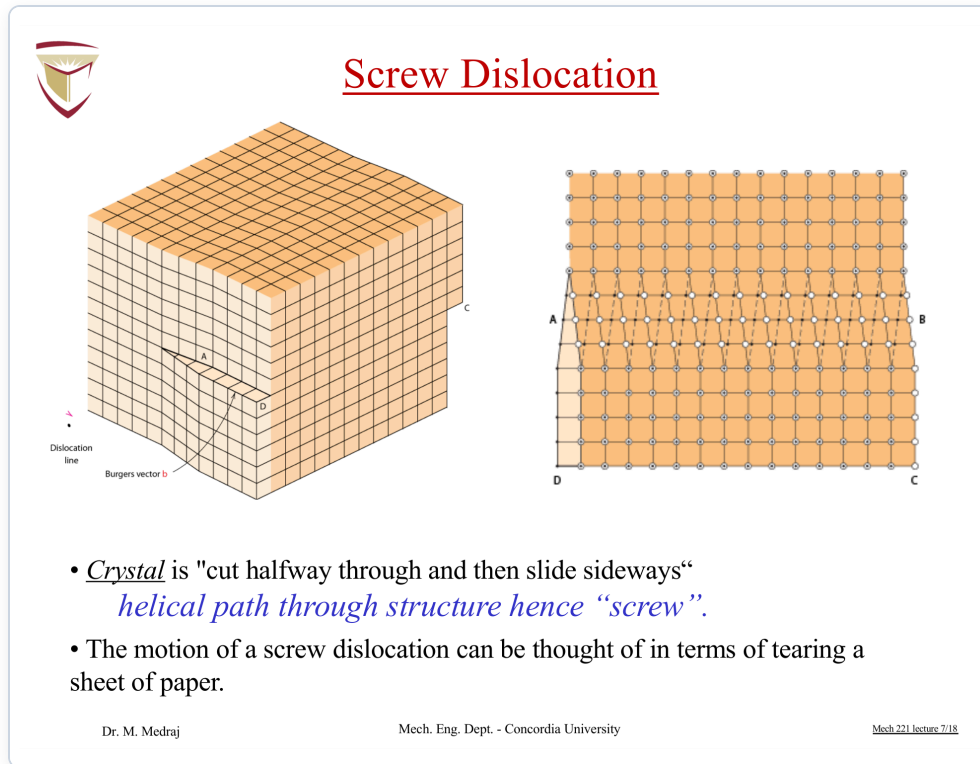


Figure 7.7: Atomic configuration of a screw dislocation (Dr. Medraj Lecture 7, Slide 18 & Callister Fig. 4.8). Formed by a shear cut and displacement, converting atomic planes into a continuous helical spiral ramp. The Burgers vector $\mathbf{b}$ is strictly parallel to the dislocation line: $\mathbf{b} \parallel \mathbf{t}$.

- **Structure:** Formed by applying a shear stress that displaces one part of the crystal by one lattice spacing relative to the other, creating a continuous helical ramp.
- **Burgers Vector Relationship:**

$$\mathbf{b} \parallel \mathbf{t} \quad (\text{Burgers vector is PARALLEL to dislocation line})$$

3. Mixed Dislocation

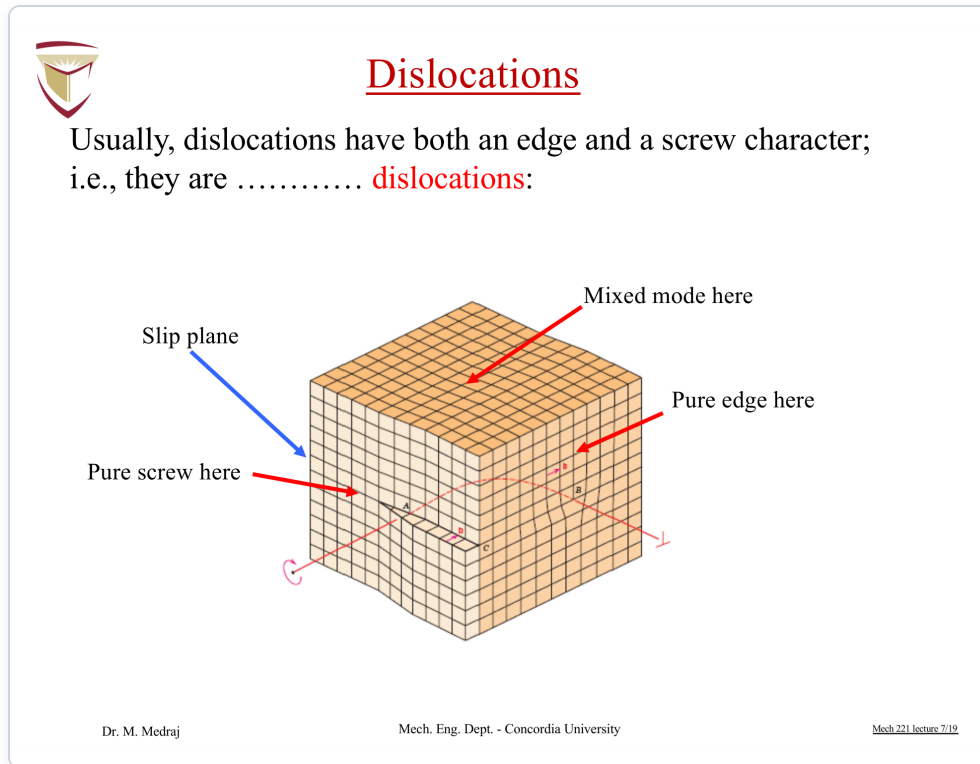


Figure 7.8: Mixed dislocation loop in a crystalline solid (Dr. Medraj Lecture 7, Slide 20 & Callister Fig. 4.9). In real engineering alloys, dislocation lines are curved; the Burgers vector $\mathbf{b}$ remains invariant across the entire line, transitioning continuously from pure edge ($\mathbf{b} \perp \mathbf{t}$) to pure screw ($\mathbf{b} \parallel \mathbf{t}$).

- Real dislocations in engineering metals are rarely purely edge or purely screw; they curve through the lattice as **mixed dislocations**.
- The **Burgers vector $\mathbf{b}$ is invariant** along the entire length of the dislocation line.
- At points where $\mathbf{b} \perp \mathbf{t}$, the character is pure edge.
- At points where $\mathbf{b} \parallel \mathbf{t}$, the character is pure screw.
- Everywhere else ($0^\circ < \theta < 90^\circ$), the character is **mixed**.

The Burgers Circuit Algorithm

The **Burgers vector ($\mathbf{b}$)** denotes the magnitude and direction of the lattice distortion created by a dislocation:

1. Traverse equal numbers of lattice steps in a closed right-handed loop around the dislocation line in the real crystal (m steps up, n steps right, m steps down, n steps left).
2. Trace the exact same sequence of steps in a perfect reference crystal.
3. The closure failure vector pointing from the finish point to the starting point in the reference crystal is the **Burgers vector $\mathbf{b}$** .

6. Master Formula & High-Yield Summary Matrix

Defect / Parameter	Governing Relationship	Physical Interpretation & Rules
Equilibrium Vacancies	$\frac{N_v}{N} = \exp\left(-\frac{Q_v}{k_B T}\right)$	Exponential increase with T ; T must be in Kelvin.
Lattice Site Density	$N = \frac{\rho N_A}{A}$	Convert ρ to g/m^3 to find sites per m^3 .
Arrhenius Slope	Slope $m = -\frac{Q_v}{k_B}$	Plot $\ln(N_v/N)$ vs $1/T$; slope yields activation energy.
Hume-Rothery Rules	1. $\Delta r < 15$, 2. Same crystal, 3. Low ΔX , 4. Valency	Dictates complete vs limited substitutional solid solubility.
Edge Dislocation	$\mathbf{b} \perp \mathbf{t}$	Extra half-plane; glide direction parallel to $\mathbf{b}$.
Screw Dislocation	$\mathbf{b} \parallel \mathbf{t}$	Helical ramp; glide direction perpendicular to $\mathbf{b}$.
Mixed Dislocation	$0^\circ < \angle(\mathbf{b}, \mathbf{t}) < 90^\circ$	Burgers vector $\mathbf{b}$ is constant; character transitions along curve.